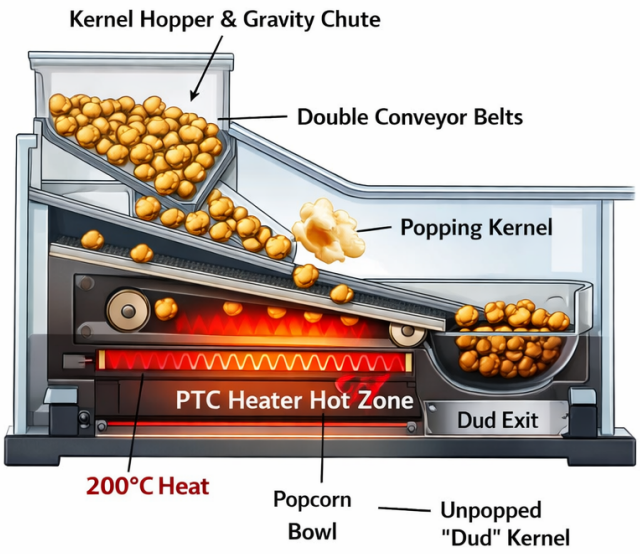
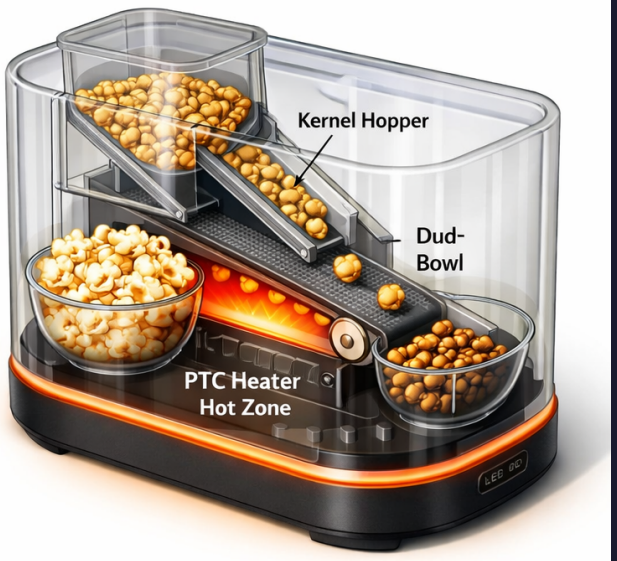
Pop!

One kernel at a time. Physics does the sorting.

A miniature desktop popcorn machine that pops kernels individually on a slow conveyor belt through a hot zone. Popped kernels escape into a bowl. Duds ride to the end. USB-C powered. Mesmerizing.





The Problem

Microwave popcorn is invisible

Push a button, wait 3 minutes, listen for the slowdown. Nothing to see. No ritual. No spectacle. The bag arrives pre-made with artificial butter flavor and regret.

Stovetop is overkill

Oil, pot, lid, constant shaking, burned kernels stuck to the bottom, cleanup. All for a snack. Nobody does this at their desk.

No desk-scale popcorn experience exists

There is nothing that turns popcorn into a meditative, kernel-by-kernel show you watch while you work. Pop! fills the space between snack and spectacle.

TARGET USERS

Desk workers | Dorm dwellers | Gadget lovers

Gift recipients | Snack enthusiasts | Ages 14+

How It Works

1 Fill

Pour kernels into the hopper

Lift the clear lid, pour in two tablespoons of kernels. The hopper holds ~100. Plug in the USB-C cable.

2 Start

Press the button, wait 45 seconds

LED ring glows amber as the PTC heater preheats to 200C. When the ring turns green, the conveyor belt starts.

3 Watch

Kernels pop one at a time

Kernels slide down the gravity chute single-file onto the belt. The belt carries each through the hot zone. Pop! The kernel explodes, escapes the belt, tumbles into the bowl.

4 Eat

Popcorn bowl fills, duds go to waste

Popped kernels escape sideways into the clear bowl. Duds ride the belt to the end and drop into the dud bowl. No sensors, no sorting logic. Physics does it.

Architecture

Kernel Pipeline

Hopper --> Chute --> Belt --> Hot Zone --> Bowl / Dud

gravity single-file slow grip 200C PTC physics sorting

Key Insight: No Sensors for Sorting

Popped = 25mm, escapes belt. Unpopped = 6mm, stays gripped. Physics wins.

SUBSYSTEMS

Double Conveyor Belt

PTFE-fiberglass mesh, grips kernels through hot zone, 5-8 mm/s

PTC Ceramic Heater (20W)

Self-regulating, 200C, mounted under belt path

Gravity Chute

Angled slide, narrows to 9mm, single-files kernels passively

MCU (ESP32-C3)

PID temp control, motor PWM, LED effects, optional BLE

USB-C PD (30W)

20V @ 1.5A via STUSB4500, no wall brick

LED Ring (WS2812B x8)

Amber = preheat, green = running, blue = done, red = fault

Dumb firmware, smart mechanism

The firmware runs PID temperature control and constant-speed motor drive. No pop detection, no sorting logic, no jam recovery. The conveyor belt and physics handle everything the firmware doesn't need to know about.

Physics-Based Sorting

How does the machine know which kernels popped?

Unpopped Kernel (Dud)

Size: ~6mm | Hard, round

Fits snugly between the two belts. Belt tension grips it. Rides the full belt path. Drops into the dud bowl at the end.

Popped Kernel

Size: ~25mm | Expanded 10-15x | Irregular

Way too big for the belt gap. The expansion force pushes the spring-loaded belts apart. The kernel breaks free and escapes sideways into the popcorn bowl. No sensor needed.

The conveyor belt IS the sensor

- **Belt gap: ~4mm at rest:**
  Spring-loaded rollers apply light tension. 6mm kernels get gripped. 25mm popcorn can't be held.
- **Belt speed: 5-8 mm/s:**
  Each kernel spends 10-16 seconds in the hot zone. Fast enough to be interesting, slow enough to pop.
- **Belt material: PTFE-coated fiberglass:**
  Survives 260C continuous. Food-safe (FDA approved). Low-stick so popcorn doesn't adhere.

Constraints & BOM

Hot zone temp — **~200C**

PTC self-regulating, PID-controlled

Belt speed — **5-8 mm/s**

10-16s per kernel in hot zone

Power — **USB-C PD 30W**

19W steady state, 11W headroom

Footprint — **150 x 180 x 140mm**

Fits on a desk next to monitor

Weight — **~400g**

Light enough to move with one hand

Food safety — **FDA / EU 1935/2004**

PTFE belt, Tritan bowls

BOM ESTIMATE (1k units)

ESP32-C3-MINI-1	$1.50
STUSB4500 PD controller	$1.20
USB-C + buck regulator	$0.90
MOSFET + DRV8837 motor driver	$1.00
PTC ceramic heater (20W)	$1.50
DC gearmotor (30 RPM)	$2.00
Conveyor belt assembly (belts + rollers + springs)	$3.50
Stainless steel chute	$0.80
NTC thermistor + passives	$0.65
WS2812B LEDs x8 + piezo	$1.00
PCB (50x40mm, 4-layer)	$1.20
Enclosure (hopper + housing + base + bowls)	$5.10
Thermal insulation	$0.30
Packaging + USB-C cable + sample kernels	$1.50
Assembly + test	$2.50
Total COGS	**~$24.75 (5k: ~$18.50)**

Hardest Problems

1 Belt grip vs. pop escape

The belt must grip unpopped kernels (~6mm) firmly enough to transport them, yet release popped kernels (~25mm) reliably. Spring-loaded belt tension is the tuning variable. Too tight: popped kernels get crushed. Too loose: unpopped kernels slip. Must prototype with real kernels and real belts at M1.

2 Heat transfer through the belt

Kernels are heated through PTFE-fiberglass mesh from a PTC heater below. If heat transfer is too slow, each kernel takes >12 seconds to pop and the cadence is boring. May need heater on both sides or direct radiant exposure through mesh openings. Target: pop within 8-12 seconds of entering the hot zone.

3 Gravity chute jamming

The passive chute must single-file kernels from a bulk hopper using only geometry and gravity. Kernels are 4-8mm, irregular, and tend to bridge in narrowing channels. V-groove cross-section and polished surface help. A small vibration motor is the fallback. Must test with multiple kernel brands at M1.

All three are mechanism problems solved by prototyping, not simulation. M1 milestone: 3D-printed frame, off-shelf belts, PTC heater, real kernels.

Gate Result & Next

GATE: PASS

70 pass / 13 N/A / 3 minor

System description complete.

Ready for mechanism prototype.

POWER

USB-C PD: 20V @ 1.5A (30W) | Steady state: ~19W

Heater 17W | Motor 0.3W | MCU 0.15W | LEDs 0.8W | 11W headroom

KEY SPECS

150x180x140mm | ~400g | Clear enclosure | Tritan bowls

ESP32-C3 | BLE 5.0 | OTA | Retail $50-80 | BOM <$25

8 OPEN ITEMS

M1	Belt grip vs. pop escape tuning with real kernels
M1	Heat transfer rate through PTFE-fiberglass belt
M1	Gravity chute jamming across kernel brands
M2	Popped kernel escape direction / bowl placement
M2	Thermal isolation (outer surface <45C)
M3	Belt longevity at 200C (target 500+ hours)
M2	UL/ETL path for <25W heating appliance
M2	USB-C PD compatibility across charger brands

Next step: 3D-print the frame, source PTFE belts and a PTC heater, build the conveyor mechanism. Pop real kernels. Validate belt grip, chute flow, and escape geometry before committing to tooling.